O-C-105c
August 11, 1965

SUPERSEDING
Int. Fed. Spec. O-C-00105b
July 15, 1963 and
Fed. Spec. O-C-105a
September 18, 1958

FEDERAL SPECIFICATION

CALCIUM CHLORIDE, DIHYDRATE AND CALCIUM CHLORIDE, ANHYDROUS; TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification is intended to establish the requirements for the technical grades of hydrated and anhydrous calcium chloride (6.1).

1.2 Classification.

1.2.1 *Types, grades, and classes.* The calcium chloride covered by this specification shall be of the following types, grades, and classes, as specified:

Type I—Dihydrate, 77 percent, flake.
Type II—Anhydrous.
  Grade A—94 percent, flake, pellet, or other granular.
  Grade B—Desiccant.
    Class 1—Granular.
    Class 2—Lump.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-S-48—Sacks, Shipping, Paper.
PPP-B-35—Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-C-186—Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals.
PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6810

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Publications:

ASTM Standards—Part 4.
D 98—Calcium Chloride.
D 345—Standard Methods of Sampling and Testing Calcium Chloride.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa.)

3. REQUIREMENTS

3.1 Material. Calcium chloride shall not be caked or sticky in the container when received and shall conform to the chemical requirements shown in table I.

3.2 Particle size. The material shall conform to the requirements shown in table II when tested as specified in 4.4.2.

TABLE I. *Chemical requirements*

	Type I	Type II		Test paragraph
		Grade A	Grade B	
Calcium chloride, minimum percent*	77.0	94.0	90.0	4.4
Total alkali chloride (as NaCl), maximum, percent	2.0	5.0	4.2	4.4
Total magnesium (as $MgCl_2$), maximum, percent	0.5	0.5	0.5	4.4
Other impurities (not including water) (maximum) percent	1.0	1.0	1.0	4.4

* When material is inspected at destination, the calcium chloride content shall be in accordance with ASTM D 98, paragraph 6.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification

TABLE II. *Sieve analysis*

Sieve size or number	Type I	Type II		
		Grade A	Grade B	
			Class 1	Class 2
Passing 1-inch sieve				100 percent
Passing 3/8 inch sieve	100 percent	100 percent	100 percent	—
Passing U. S. No. 4	80 percent, minimum	80 percent, minimum	90 percent, minimum	5 percent maximum
Passing U. S. No. 8	—	—	50 percent, maximum	—
Passing U. S. No. 16	—	—	5 percent maximum	—
Passing U. S. No. 20	10 percent, maximum	10 percent, maximum	—	—

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where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type, grade, and class submitted for inspection and delivery at one time.

4.3 Inspection.

4.3.1 *Inspection of material and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 *Inspection and test of the end item.*

4.3.2.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of material of the same type and class for examinations under 4.3.2.1.1, 4.3.2.1.2, and 4.3.2.1.3, and in units of shipping containers for examination under 4.3.2.1.5.

4.3.2.1.1 *Examination of the end item container for defects in closure, appearance, markings, construction, and workmanship.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Container	Not specified type, size or style container.
	Any evidence of leakage or seepage from any part of the container.
	Any hole, tear, puncture, scuff, dent or other defect in body of container.
	Closures not as specified.
Labeling	Printing or stenciling pertaining to the applicable information and instructions for end item use of the specified type material omitted, illegible, incorrect, incomplete, smeared.

4.3.2.1.2 *Examination of the end item container for defects in appearance of material.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Appearance of calcium chloride	Presence of dirt or foreign matter.
	Not homogeneous.
	Form not as specified.
	Objectionable odor.

4.3.2.1.3 *Examination of the end item container for defects in net contents.* The sample unit for this examination shall be one filled container. The average net contents shall be not less than the specified or indicated quantity.

4.3.2.1.4 *Testing of the end item.* From each inspection lot, two containers shall be selected at random. From each of the two containers enough material for tests shall be taken and placed in separate, clean, dry glass containers, sealed, marked, and forwarded to the testing laboratory designated by the activity concerned.

4.3.2.1.4.1 Each of the containers shall be sampled by scraping aside the top layer to a depth of approximately 1-inch and taking samples by means of a sampling thief or other method which will insure a sample that is representative of the material in the container to a depth of at least six inches. Precautions should be taken during the sampling operation to avoid exposing the sample unduly to atmospheric moisture.

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4.3.2.1.5 *Examination of preparation for delivery.* An examination will be made to determine that markings, materials, workmanship, contents, and weights comply with the requirements of section 5. Defects will be scored in accordance with the classification below. The sample unit will be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. This examination does not apply when the primary container is the shipping container.

Examine	Defect
Packing (as applicable)	Not in accordance with contract requirements. Not level specified. Inadequate application of components such as incomplete closures of containers, bulged or distorted drums.
Packing	Container not as specified; closures not accomplished by specified or required methods or materials. Any nonconforming component, component missing, damaged, or otherwise defective.
Weight	Gross or net weight exceeds specified requirements.
Markings	Omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

4.3.2.1.6 *Inspection levels and AQLs for examinations.* The inspection level for determining the sample size, and the AQLs expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2.1.1	S-3	2.5
4.3.2.1.2	S-3	1.5
4.3.2.1.3	S-2	—
4.3.2.1.5	S-2	4.0

4.4 Test procedures.

4.4.1 *Tests.* Tests shall be made in accordance with the applicable method of ASTM D 345 and as specified hereinafter.

4.4.2 *Particle size.* Sieve tests shall be made in accordance with applicable sections of method 2101 of Fed. Test Method Std. No. 536. The sieves being used shall be nested with pan and top, and sieving shall be completed as rapidly as possible.

5. PREPARATION FOR DELIVERY

For civil agency requirements, the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.3).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Calcium chloride shall be packaged in one-pound or five-pound quantities as specified (see 6.3), in glass bottles conforming to group A, class 1, style 2, closure A of PPP-C-186.

5.1.1.2 *Intermediate packaging.* Calcium chloride unit packaged in accordance with 5.1.1.1 shall be intermediately packaged in accordance with the level A requirements of PPP-C-186.

5.1.2 *Level C.* Calcium chloride shall be packaged in the quantities specified in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.3).

5.2.1 *Level A.*

5.2.1.1 *One- and five-pound quantities.* Calcium chloride packaged in accordance with 5.1.1 shall be packed in accordance with the level A requirements of PPP-C-186.

5.2.1.2 *Eighty- and one hundred-pound quantities.* Calcium chloride in eighty- or one hundred-pound quantities, as specified (see 6.3), shall be packed in appropriate size sacks conforming to construction num-

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ber 16X or 17X of UU-S-48, bags conforming to bag number P35B/A of PPP-B-35, or fiber drums conforming to type III, grade A of PPP-D-723, as specified (see 6.3). Fiber drums shall be provided with a polyethylene bag liner or other suitable water vaporproof liner or interior coating.

5.2.2 *Level B.*

5.2.2.1 *One- and five-pound quantities.* Calcium chloride in one- and five-pound quantities packaged in accordance with 5.1.1, shall be packed in accordance with the level B requirements of PPP-C-186.

5.2.2.2 *Eighty- and one hundred-pound quantities.* Calcium chloride in eighty- or one hundred-pound quantities as specified (see 6.3), shall be packed in appropriate size sacks conforming to construction number 8 L/W or 14 L/W of UU-S-48, bags conforming to bag number P35B of PPP-B-35, or fiber drums conforming to type I, grade A of PPP-D-723, as specified (see 6.3). Drums shall be provided with a polyethylene bag liner or other water vaporproof lining or interior coating.

5.2.3 *Level C.* Calcium chloride shall be packed in the quantities specified in a manner which will insure carrier acceptance at lowest rates and safe delivery at destination. Containers and packing shall comply with rules or regulations of carriers as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended uses.** Type I, and type II, grade A calcium chloride are intended for use for road purposes, as admixtures for concrete, as refrigerating mediums and other technical uses. In addition, both types may be used for snow and ice control on sidewalks and driveways of buildings. Type II grade B calcium chloride is intended for use as a desiccating (drying) agent. Calcium chloride should not be used as an antifreeze material in the watercooling system of internal combustion engines. The requirements for type I and type II, grade A calcium chloride are essentially similar to the requirements of ASTM D 98.

6.2 **CAUTION:** Calcium chloride is a very hydroscopic material and should be stored in tightly closed moisture-resistant containers.

6.3 **Ordering data.** Purchasers should exercise any desired options offered herein and should include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type, grade, and class of material required (1.2.1).

(*c*) Unit quantities and levels of packaging and packing required (5.1 and 5.2).

(*d*) Purchasers desiring bags of calcium chloride are advised that normally industry bags type I in 100 pound multiwall paper sacks and type II grade A in 80 pound multiwall paper sacks.
(Both types are also furnished in 100 lb. fiber drums.)

6.4 **Standard pack for civil agencies.** The standard pack requirements shall be 100 lb. paper sacks for type I and 80 lb. paper sacks for type II, grade A, as specified in 5.2.2.2, and are intended for use in procurement of stores stock replenishments. Procurement officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot

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or intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.

6.5 Level B packaging. When level B packaging is specified for civil agency procurement the level A requirements in 5.1 shall apply.

MILTARY CUSTODIANS:

Army—MU
Navy—YD
Air Force—68

Review activities:

Army—MU, MR
Navy—YD, SH
Air Force—68

User activities:

Navy—MC

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